

AIR QUALITY INDEX (AQI) ANALYSIS OF DELHI

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Domain: Data Science

Organization: Shadow-Fox Virtual Internship Program

INTRODUCTION

Air pollution has become one of the most significant environmental and public health concerns worldwide. Delhi, the capital city of India, is frequently ranked among the most polluted cities globally due to rapid urbanization, industrial activities, vehicular emissions, and seasonal factors. The Air Quality Index (AQI) is used to measure and communicate air quality levels and their potential impact on human health.

This project aims to analyse Delhi's air quality data by examining various pollutants, identifying pollution trends, determining peak pollution periods, and understanding relationships between different pollutants through statistical analysis and visualization techniques.

Objectives

The primary objectives of this project are:

- To analyse the concentration of major air pollutants in Delhi.
- To identify periods of high and low pollution levels.

- To determine the pollutants contributing most significantly to air pollution.
- To study the relationship between different pollutants.
- To derive meaningful insights that can support environmental monitoring and public health initiatives.

Dataset Description

The dataset consists of hourly air quality observations collected from Delhi.

```
Shape: (561, 9)
```

```
Columns:
```

```
Index(['date', 'co', 'no', 'no2', 'o3', 'so2', 'pm2_5', 'pm10', 'nh3'], dtype='object')
```

	date	co	no	no2	o3	so2	pm2_5	pm10	nh3
0	2023-01-01 00:00:00	1655.58	1.66	39.41	5.90	17.88	169.29	194.64	5.83
1	2023-01-01 01:00:00	1869.20	6.82	42.16	1.99	22.17	182.84	211.08	7.66
2	2023-01-01 02:00:00	2510.07	27.72	43.87	0.02	30.04	220.25	260.68	11.40
3	2023-01-01 03:00:00	3150.94	55.43	44.55	0.85	35.76	252.90	304.12	13.55
4	2023-01-01 04:00:00	3471.37	68.84	45.24	5.45	39.10	266.36	322.80	14.19

Dataset Information

- Total Records: 561
- Total Columns: 9
- Time Period: January 2023
- Observation Frequency: Hourly

Features Included

Feature Description

date	Date and time of observation
co	Carbon Monoxide concentration
no	Nitric Oxide concentration
no2	Nitrogen Dioxide concentration
o3	Ozone concentration
so2	Sulfur Dioxide concentration
pm2_5	Particulate Matter (PM2.5)
pm10	Particulate Matter (PM10)
nh3	Ammonia concentration

Data Preprocessing

Before performing the analysis, the dataset was cleaned and prepared using Python and Pandas.

Steps Performed

1. Loaded the dataset using Pandas.
2. Inspected the structure and dimensions of the dataset.
3. Checked for missing values.
4. Checked for duplicate records.
5. Converted the date column into datetime format.
6. Extracted Hour, Day, and Month features for time-based analysis.
7. Calculated an overall AQI score using the available pollutant values.
8. Categorized AQI levels into different air quality categories.

```

RangeIndex: 561 entries, 0 to 560
Data columns (total 9 columns):
..  #      Column  Non-Null Count  Dtype
   ---  -
    0    date      561 non-null    object
    1    co         561 non-null    float64
    2    no         561 non-null    float64
    3    no2        561 non-null    float64
    4    o3         561 non-null    float64
    5    so2        561 non-null    float64
    6    pm2_5      561 non-null    float64
    7    pm10       561 non-null    float64
    8    nh3        561 non-null    float64
dtypes: float64(8), object(1)
memory usage: 39.6+ KB
None

Missing Values:
date      0
co         0
no         0
no2        0
o3         0
so2        0
pm2_5      0
pm10       0
nh3        0
dtype: int64

```

Data Quality Assessment

- Missing Values: 0
- Duplicate Records: 0

The dataset was found to be clean and suitable for analysis.

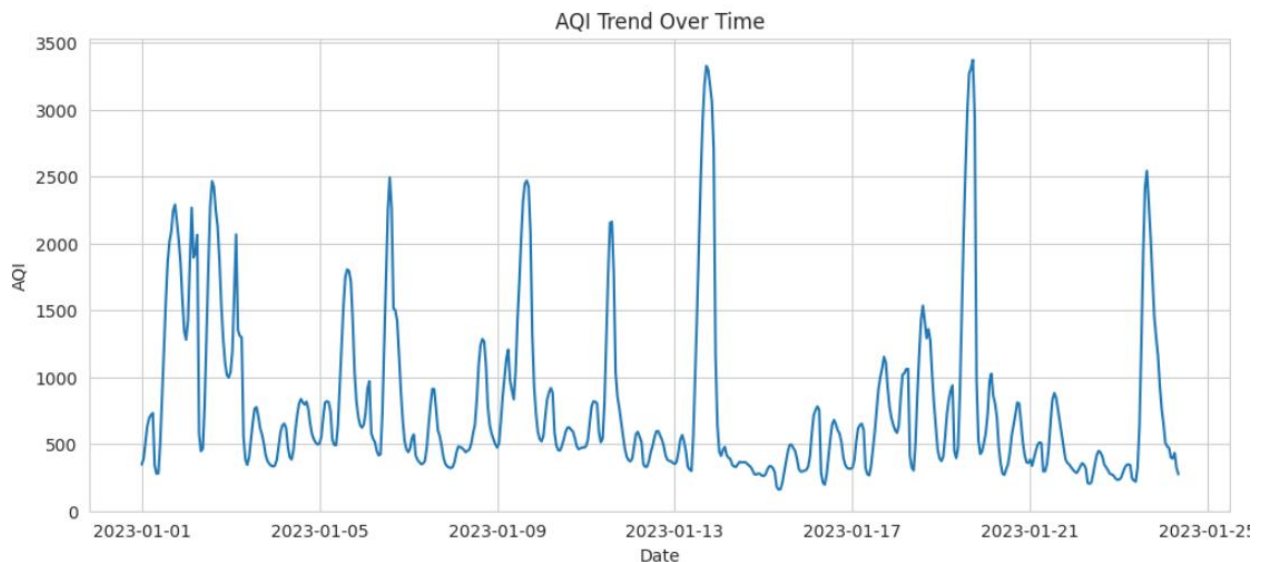
Visualization and Analysis

1. AQI Trend Over Time

The AQI trend graph illustrates how air pollution varied throughout the observation period.

Observations

- AQI levels fluctuated significantly across different days.
- Several pollution spikes were observed during the study period.
- The highest AQI values were recorded during mid and late January.
- Pollution levels remained consistently high during multiple periods.

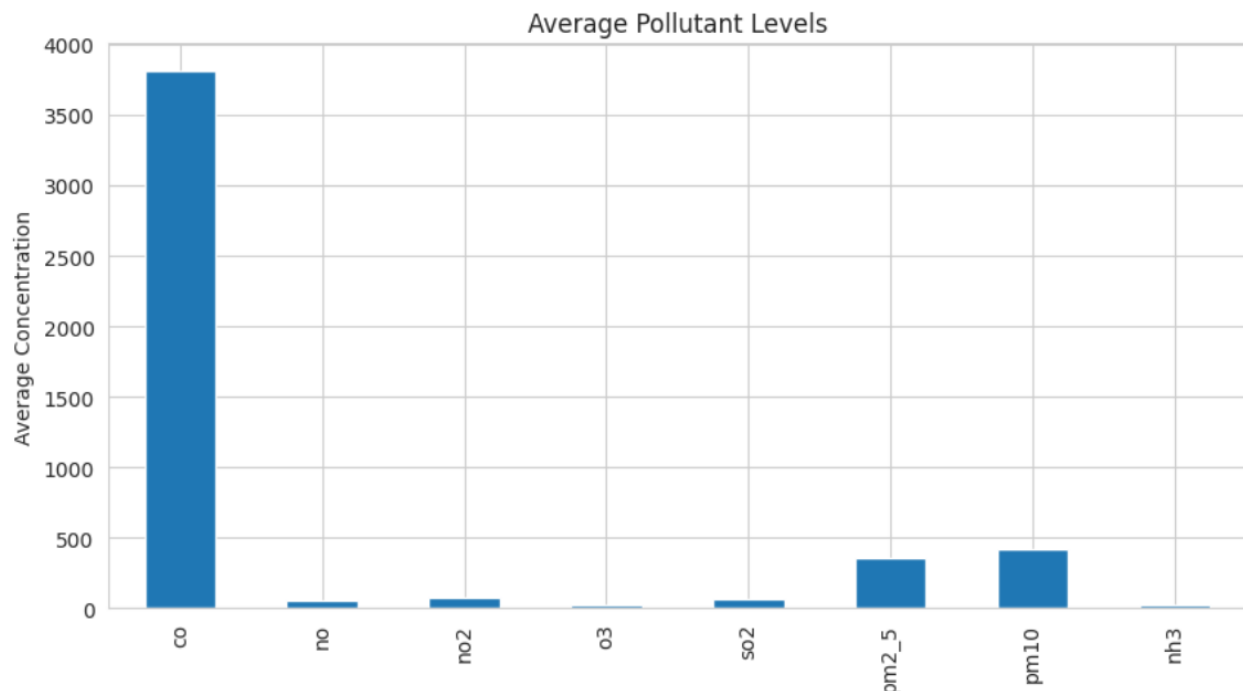


2. Average Pollutant Levels

The average pollutant comparison chart was used to determine which pollutants contributed most significantly to air pollution.

Observations

- Carbon Monoxide (CO) recorded the highest average concentration.
- PM10 and PM2.5 were among the most dominant particulate pollutants.
- Ozone (O₃) showed comparatively lower average concentrations.
- Particulate matter contributed significantly to poor air quality conditions.



3. AQI Category Distribution

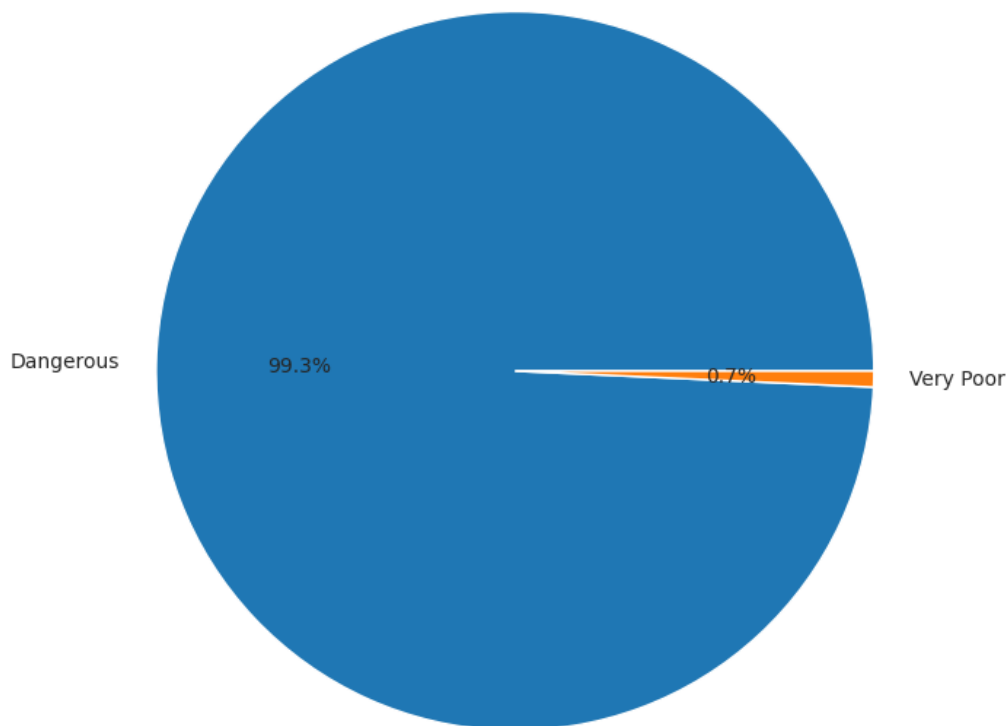
AQI values were categorized into air quality classes to assess the overall severity of pollution.

Observations

- A large majority of observations fell under the Dangerous category.
- Very few observations were classified as Very Poor.
- No observations were found in the Good or Very Good categories.

This indicates persistently poor air quality throughout the study period.

AQI Category Distribution

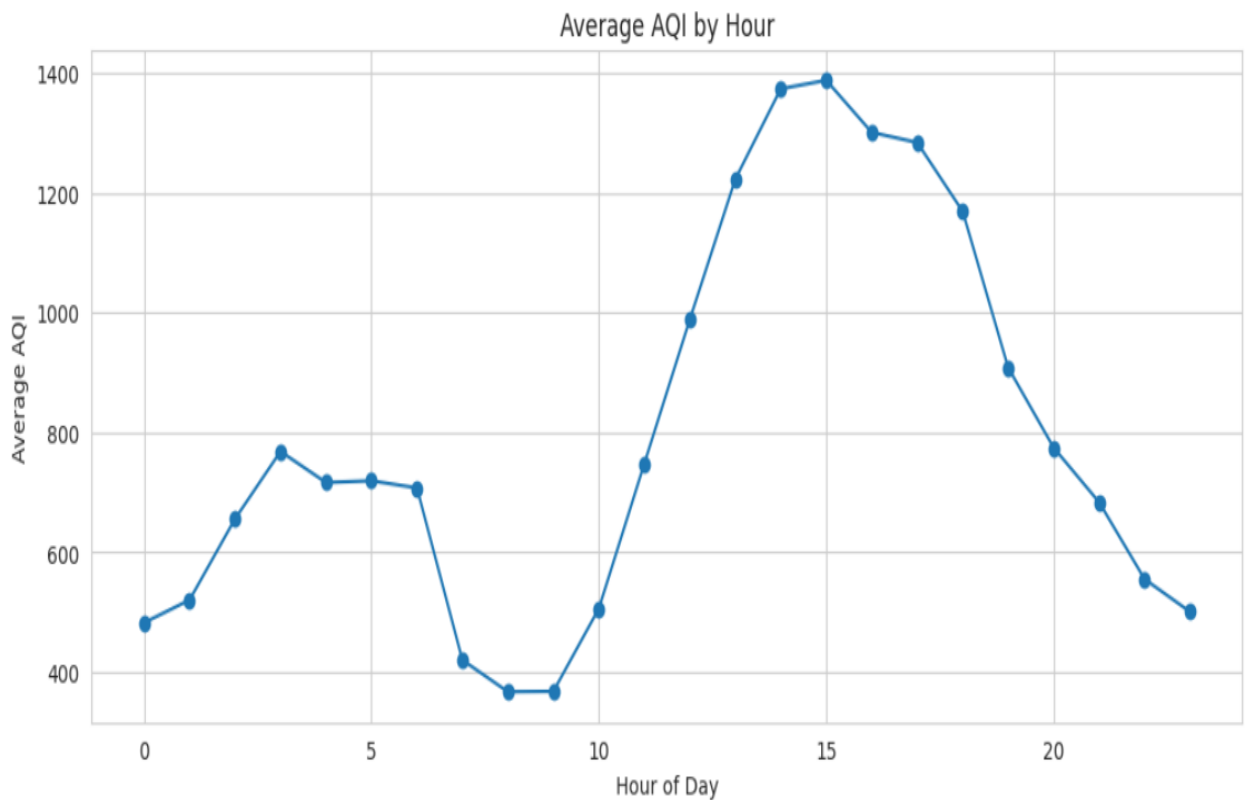


4. Average AQI by Hour

Hourly AQI analysis was performed to identify peak pollution periods during the day.

Observations

- AQI values increased significantly during afternoon hours.
- Peak pollution levels were observed between 2 PM and 3 PM.
- Lower AQI levels were observed during the early morning hours.
- Air quality varied considerably throughout the day.



5. Top Polluted Records

The top 10 highest AQI observations were identified.

Observations

- The highest AQI value was recorded on January 19, 2023.
- Multiple extreme pollution events occurred on January 13 and January 19.
- These periods represent the most critical air quality conditions in the dataset.

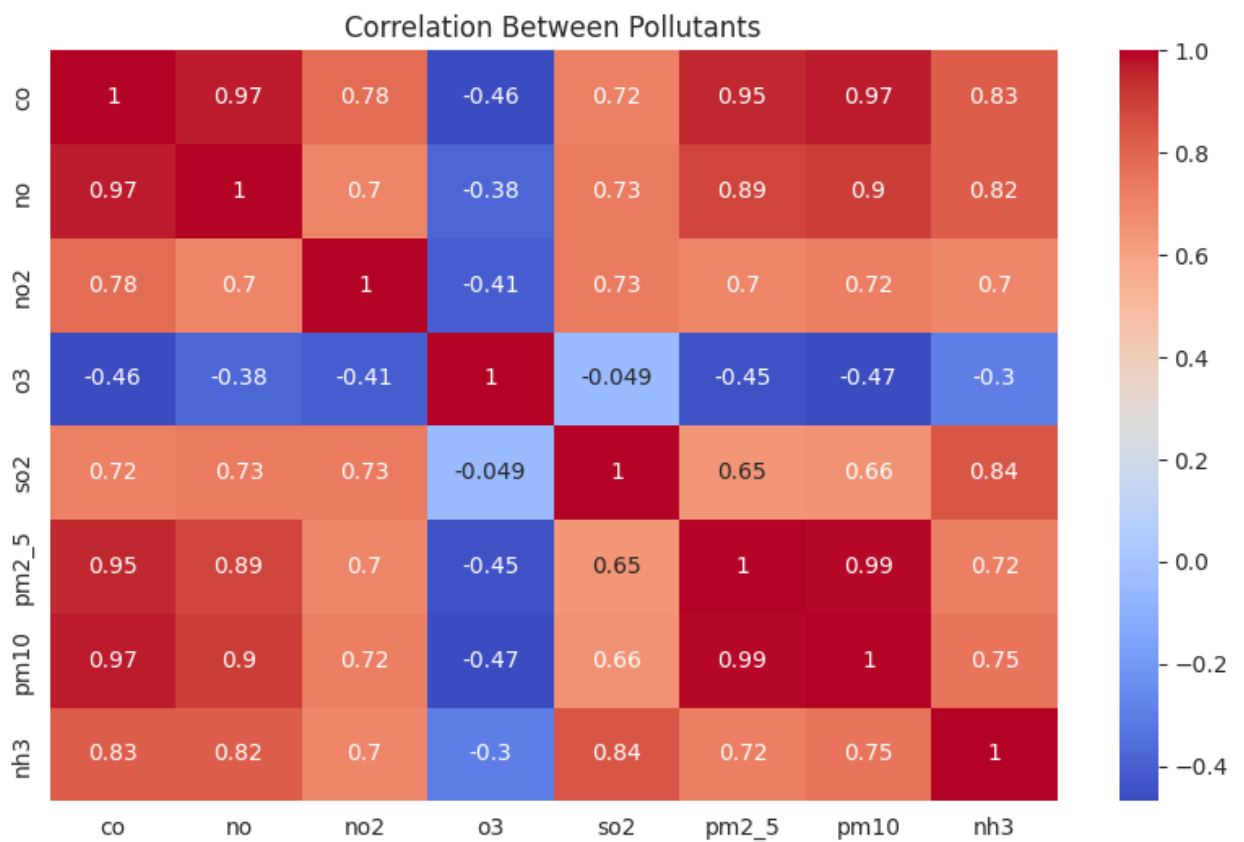
	date	AQI
449	2023-01-19 17:00:00	3372.581667
305	2023-01-13 17:00:00	3327.930000
448	2023-01-19 16:00:00	3308.568333
306	2023-01-13 18:00:00	3299.521667
447	2023-01-19 15:00:00	3272.543333
307	2023-01-13 19:00:00	3185.583333
304	2023-01-13 16:00:00	3182.850000
308	2023-01-13 20:00:00	3062.546667
446	2023-01-19 14:00:00	2987.196667
450	2023-01-19 18:00:00	2940.668333

6. Correlation Analysis

A correlation heatmap was generated to understand relationships between pollutants.

Observations

- PM2.5 and PM10 exhibited a very strong positive correlation (0.99).
- CO showed a strong positive relationship with PM10 and PM2.5.
- NO and NO2 displayed moderate to strong correlations.
- Ozone (O3) showed negative correlations with several pollutants.



Interpretation

The strong correlation between PM_{2.5} and PM₁₀ suggests that particulate pollutants often increase and decrease together, indicating common pollution sources and atmospheric behavior.

Key Findings

- Carbon Monoxide (CO) was the dominant pollutant in the dataset.
- PM_{2.5} and PM₁₀ contributed significantly to poor air quality.
- Peak AQI values occurred during afternoon hours.
- January 19 recorded some of the highest pollution levels.
- PM_{2.5} and PM₁₀ exhibited an extremely strong positive correlation.
- Ozone displayed negative relationships with several major pollutants.

Conclusion

An in-depth analysis of Delhi's Air Quality Index (AQI) dataset was conducted using Python, Pandas, Matplotlib, and Seaborn. The analysis revealed consistently poor air quality conditions, with Carbon Monoxide (CO), PM10, and PM2.5 emerging as the dominant pollutants. AQI levels were found to peak during afternoon hours, particularly between 2 PM and 3 PM. Correlation analysis showed a very strong relationship between PM2.5 and PM10, indicating that particulate pollution plays a major role in Delhi's air quality. These findings highlight the importance of continuous environmental monitoring and the implementation of effective pollution control measures to improve public health and environmental sustainability.